Software Requirements Specification

For

PathPilot

(Web-Based Career Guidance, Learning & Content Creation Platform)

Version 1.0 approved

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<22-Mar-2026>

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Revision History

Name	Date	Reason For Changes	Version
Ram Prakash Kurmi, Maheshwar Chaudhary & Gautam Tharu	22-Mar-2026	Initial Document Creation	1.0

1. Introduction

1.1 Purpose

This document presents the Software Requirements Specification (SRS) for PathPilot, a web-based career guidance, structured learning, and user-generated content platform. The purpose of this document is to provide a complete and unambiguous description of the functional and non-functional requirements of the PathPilot system. It serves as a formal agreement between the development team, academic stakeholders, and project evaluators regarding what the system shall do and how it shall perform under defined conditions.

PathPilot serves three distinct user roles — Guest (unauthenticated visitor), Learner/Creator (logged-in user who can both learn and publish career paths), and Administrator — each with a clearly defined and escalating set of capabilities. The platform enables creators to build and publish career roadmaps, monitor how many learners are enrolled in their paths, and moderate their learner communities by banning disruptive users from their content.

1.2 Document Conventions

This document conforms to the IEEE Std 830-1998 recommended practice for Software Requirements Specifications. The following conventions are applied consistently throughout:

- Document Title Page: Contains the project title, author details, institution name, version number, and submission date.
- Table of Contents: Provides a complete, numbered, and page-referenced listing of all sections and subsections.
- Section Headings: Numbered hierarchically (e.g., 1, 1.1, 1.1.1) and formatted in descending font size to indicate document hierarchy.
- Bold Text: Reserved for key terms, critical requirements, and section labels requiring special attention.
- Bullet Points: Used to enumerate system features, constraints, user roles, and requirements for improved readability.
- Shall vs. Should: The word 'shall' denotes a mandatory requirement; 'should' denotes a recommended but non-mandatory requirement.
- Database Schema: Defines tables such as Users, CareerPaths, Phases, Modules, Enrollments, Progress, PathBans, and Certificates along with their attributes and relationships.
- UI Design Reference: Describes screen layouts and navigation flows based on the PathPilot Figma prototype (Version 1.0, March 2026).

1.3 Intended Audience and Reading Suggestions

The intended audience for this SRS document encompasses all stakeholders involved in the development, testing, deployment, and evaluation of the PathPilot platform. The following table summarises reading recommendations by role:

Audience	Recommended Sections
Developers	Sections 3, 4, 5 — System Features, Interfaces, Non-Functional Requirements
Testers / QA	Sections 3, 5 — Features and Performance / Security benchmarks
Admin / Moderators	Sections 2.3, 3.7 — User Classes and Admin Dashboard
Learners / End Users	Sections 2.2, 3.4, 3.5, 3.6 — Features, Enrollment, Modules, Certificates
Content Creators	Sections 2.3, 3.2, 3.3 — Creator role, Career Path Management, Learner Analytics
Academic Evaluators	Entire document

1.4 Project Scope

PathPilot is a full-featured, web-based career guidance and learning platform designed to serve three primary user roles: Guest, Learner/Creator, and Administrator. The system scope is defined as follows:

- A publicly accessible web platform enabling guests to browse career paths, view platform features, explore resources, and read about the platform before registering.
- A secure user registration and login system with email verification and role-based access control (RBAC). All registered users share the single USER role and are simultaneously both Learners (can enrol in any path) and Creators (can build and publish their own career paths).
- A structured career roadmap authoring module where any logged-in user (Creator) can build multi-phase learning paths with associated modules and resources, which are then publicly available for other users to enrol in.
- A Creator Analytics Dashboard enabling the creator of a path to view the total number of learners enrolled in each of their published paths, giving visibility into content reach and engagement.
- A Creator Content Moderation capability allowing the creator of a path to ban specific learners from their content — revoking that learner's access to the path — without requiring admin intervention.
- An enrollment and progress tracking system enabling students to enrol in career paths (including those created by other users or by admins), track completion of individual modules phase-by-phase, and monitor their learning journey from a personalised dashboard.
- A module content delivery interface allowing students to view and consume learning materials, continue from previously completed phases, and mark progress on each module.
- A certificate generation module that awards students with verifiable completion certificates upon finishing all phases of an enrolled career path.
- A resource library accessible to both guests and logged-in users, featuring curated educational materials relevant to the available career paths.
- A comprehensive Admin Dashboard providing full platform analytics, global user account management (including role changes and account suspension), career path oversight

across all creators, platform content management (Home, Features, About, Contact, Footer sections), and a notification broadcast system.

- Secure authentication mechanisms including email verification tokens, encrypted session management, and protection against common web vulnerabilities.

1.5 References

- IEEE Std 830-1998 — IEEE Recommended Practice for Software Requirements Specifications. IEEE Computer Society, 1998.
- PathPilot Figma UI Design Prototype, Version 1.0, March 2026. Prepared by Ram Prakash Kurmi, Maheshwar Chaudhary & Gautam Tharu.
- Spring Framework 6.1.6 Official Documentation. Available at: https://docs.spring.io/spring-framework/
- MySQL 8.0 Reference Manual. Available at: https://dev.mysql.com/doc/
- Jakarta EE 10 Servlet API Documentation. Available at: https://jakarta.ee/specifications/servlet/6.0/
- Tailwind CSS Documentation. Available at: https://tailwindcss.com/docs
- R.K. University B.Tech 3rd Year Project Guidelines, Academic Year 2025-2026.

2. Overall Description

2.1 Product Perspective

PathPilot is an independent, standalone web application accessed entirely through a standard web browser. The system operates on a three-tier client-server architecture comprising a responsive frontend (Tailwind CSS + JSP), a Spring MVC Java application server, and a MySQL relational database. What distinguishes PathPilot from a traditional LMS is its dual-role model: every registered user is simultaneously a Learner (who can enrol in any published career path) and a Creator (who can build and publish their own career paths, monitor enrolled learners, and moderate their community).

Administrators sit above this model and have platform-wide authority: they can manage all users, oversee all career paths regardless of creator, and control all public-facing platform content. The system communicates with a Jakarta Mail SMTP server for transactional email verification and is packaged and deployed as a WAR archive on Apache Tomcat.

2.2 Product Features

- User Registration and Login: Secure account creation with email verification via a UUID token link. Role-based login supporting Learner/Creator (USER) and Admin (ADMIN) roles with Spring MVC session management.
- Career Path Browsing: Both guests and logged-in users can browse all published career roadmaps with descriptions, phase counts, and preview details.
- Career Path Authoring (Creator): Any logged-in user can create, edit, and delete their own multi-phase career paths with associated modules and resources, making them available to the wider learning community.
- Learner Count Analytics (Creator): The creator of a career path can view the total count of learners currently enrolled in each of their published paths from their Creator Dashboard.
- Content-Level Learner Ban (Creator): The creator of a career path can ban a specific learner from their path, revoking that learner's access to the path content.
- Career Path Enrollment (Learner): Authenticated users can enrol in any published career path, including paths authored by other users or by administrators.
- Phase & Module Progress Tracking: Learners track completion of each module within their enrolled paths, with a dedicated progress view.
- Module Content Delivery: Learners access individual module content pages parameterised by phase and title.
- Certificate Generation: Learners who complete all phases and modules of an enrolled path receive a verifiable completion certificate.
- Resource Library: Curated educational materials accessible to guests and all logged-in users.
- Admin Dashboard: Platform-wide analytics, global user management, full career path oversight, platform content management for all public pages, and a broadcast notification system.

2.3 User Classes and Characteristics

Capability	Guest	Learner / Creator	Admin
Browse career paths & resources	✓	✓	✓
Register / Login	✓ (register)	✓	✓
Enrol in career paths	✗	✓	✓
Track progress & earn certificate	✗	✓	✓
Create & publish career paths	✗	✓	✓
View learner count on own paths	✗	✓ (own paths)	✓ (all paths)
Ban learner from own path	✗	✓ (own paths)	✓ (all paths)
Manage all users platform-wide	✗	✗	✓
Manage platform content (CMS)	✗	✗	✓
Broadcast notifications	✗	✗	✓

Guest

- Browse the PathPilot home page, features overview, about page, and contact page without authentication.
- Browse and preview all published career paths and roadmaps with high-level phase and module information.
- Access the public resource library and view available materials.
- Prompted to register or log in to enrol in career paths, track progress, or create content.

Learner / Creator (Logged-in User)

- All capabilities of a Guest, plus full authenticated access to the platform.
- As a Learner: Enrol in any published career path (including paths authored by other users or admins), track module-by-module progress, access module content pages, and earn completion certificates.
- As a Creator: Create, edit, and delete their own career paths with structured multi-phase roadmaps and module content. Published paths are discoverable and enrollable by all other users.
- As a Creator: Access a Creator Analytics view showing the total number of learners enrolled in each of their own paths.
- As a Creator: Ban specific learners from any of their own paths, revoking that learner's access to the path and its modules.

- Manage personal profile including name, bio, and account settings.

Administrator

- All capabilities of a Learner/Creator, plus full platform-wide authority.
- Access the Admin Dashboard with platform-wide analytics: total users, all active career paths, total enrollments, and system activity.
- Create, edit, and delete any career path on the platform — including paths owned by other users.
- Manage all registered user accounts: add new users, edit user details and roles, deactivate or suspend accounts, and delete accounts.
- Manage dynamic content for all public-facing platform sections: Home, Features, About, Contact, and Footer pages.
- Broadcast notifications to all platform users via the Admin Notify interface.
- View learner counts and moderation logs across all career paths regardless of creator.

2.4 Operating Environment

Component	Specification
Frontend	HTML5, CSS3 (Tailwind CSS CDN), JavaScript (ES6+), JSP (Jakarta Server Pages)
Backend	Java 21, Spring MVC 6.1.6, Spring JDBC (JdbcTemplate)
Application Server	Apache Tomcat 10.x (WAR deployment)
Database	MySQL 8.0 (via MySQL Connector/J 8.3.0)
Email Service	Jakarta Mail 2.0.2 (SMTP-based email verification)
Build Tool	Apache Maven (WAR packaging)
Development IDE	Eclipse IDE for Enterprise Java Developers
Operating System	Windows 10/11 (development), Linux Ubuntu (production)
Browser Support	Chrome 90+, Firefox 88+, Edge 90+, Safari 14+
Device Support	Desktop, Laptop, Tablet, Smartphone (responsive via Tailwind CSS)

2.5 Design and Implementation Constraints

- The system requires a stable internet connection for all features; Tailwind CSS is loaded from CDN and email verification depends on SMTP server availability.
- The platform is limited to web-based access; no native mobile application is included in Version 1.0.
- All registered users share the same database role (USER). The distinction between Learner and Creator behaviours is behavioural — every USER can both learn and create. A separate CREATOR database role is not required.

- Creator-level banning is path-scoped — a creator can only ban learners from their own paths. Platform-wide account suspension is exclusively an admin capability.
- The Spring MVC application is packaged as a WAR file and requires a Jakarta EE 10 compatible servlet container (Apache Tomcat 10.x or equivalent).
- Java 21 is the minimum required JDK version due to Spring Framework 6.1.6 compatibility requirements.
- The email verification system depends on a configured SMTP server. Until email verification is complete, user accounts remain inactive and login is blocked.
- All admin-side and creator-side access controls are enforced via server-side session checks; client-side restrictions alone are not relied upon.

2.6 User Documentation

- Guest / New User Guide: Browsing career paths, viewing resources, registration process, and email verification instructions.
- Learner Guide: Login, browsing and enrolling in career paths, tracking progress, accessing module content, viewing and downloading certificates, and managing profile settings.
- Creator Guide: Creating and publishing career paths with multi-phase roadmap structures, editing and deleting own paths, accessing Creator Analytics to monitor learner counts, and banning learners from specific paths.
- Admin Guide: Admin Dashboard navigation, global user account management, career path oversight across all creators, platform content management (Home, Features, About, Contact, Footer), and broadcasting notifications.
- Developer / Technical Guide: Spring MVC project structure, database setup, email service configuration, and WAR deployment on Tomcat.

2.7 Assumptions and Dependencies

- All users are assumed to have access to a modern web browser and a stable internet connection.
- The hosting server is assumed to run Apache Tomcat 10.x with Java 21 and MySQL 8.0, correctly configured and operational.
- Email verification functionality depends on a correctly configured SMTP mail server. Students cannot log in until their email address is verified.
- Career path content is assumed to be created by registered users acting as Creators; administrators may also author paths. The system does not auto-generate career content.
- The Creator ban feature is assumed to operate at the path level only. A banned learner retains their platform account and can continue enrolling in other paths.
- Database backups are assumed to be managed by the server administrator; automated backup functionality is not included in Version 1.0.
- Continuous testing, user feedback collection, and iterative improvements are planned for post-launch phases.

3. System Features

3.1 User Authentication & Management

3.1.1 Description and Priority

The User Authentication module handles all account lifecycle operations including registration, email verification, login, session management, and logout. Priority: High.

3.1.2 Functional Requirements

- REQ-AU-01: The system shall allow new users to register by providing Full Name, Email Address, and Password. Upon successful submission, a UUID-based verification token shall be generated and a verification email dispatched via SMTP.
- REQ-AU-02: The system shall prevent login by accounts whose email address has not been verified. Attempting to log in without verification shall display an appropriate error message.
- REQ-AU-03: The system shall authenticate users via email and password matched against the MySQL database. Upon successful authentication, the user's email and role (USER or ADMIN) shall be stored in the server-side Spring session.
- REQ-AU-04: The system shall redirect authenticated users to their role-appropriate dashboard: ADMIN to /admin/dashboard and USER to /user/UserHome.
- REQ-AU-05: The system shall maintain secure server-side Spring sessions post-login, with session invalidation upon logout via GET /logout.
- REQ-AU-06: Administrators shall be able to add new user accounts, edit user details and assigned roles, and deactivate or delete user accounts via the Admin user management interfaces.

3.2 Career Path & Roadmap Management (Creator)

3.2.1 Description and Priority

This module provides complete lifecycle management of career paths and learning roadmaps for any logged-in user acting as a Creator. Priority: High.

3.2.2 Functional Requirements

- REQ-CP-01: Any authenticated USER shall be able to create new career paths by specifying a path title, description, category/icon, and defining the multi-phase roadmap structure with individual module entries.
- REQ-CP-02: A Creator's path management interface shall display all career paths owned by that user, showing title, description, phase count, learner count, and action controls (Edit, Delete).
- REQ-CP-03: A Creator shall be able to edit any career path they own, including the path title, description, and all associated phase and module definitions.
- REQ-CP-04: A Creator shall be able to delete any career path they own. A confirmation prompt shall appear before permanent deletion. Upon deletion, all associated enrollments, progress records, and path bans shall be removed.
- REQ-CP-05: Both guest users and authenticated learners shall be able to browse all published career paths created by any user or admin.

- REQ-CP-06: Administrators shall be able to create, edit, and delete any career path on the platform regardless of creator ownership, via the Admin career path management interfaces (/admin/career-path, /admin/create-path, /admin/edit-path).

3.3 Learner Analytics & Content Moderation (Creator)

3.3.1 Description and Priority

This module gives Creators visibility into learner engagement on their published paths and the ability to moderate their learner community at the path level. Priority: High.

3.3.2 Functional Requirements

- REQ-LC-01: For each career path they own, a Creator shall be able to view the total count of learners currently enrolled in that path from a Creator Analytics/Dashboard view.
- REQ-LC-02: The Creator Analytics view shall display, for each owned path, the path title, total enrolled learner count, active learner count, and a list of enrolled learners (display name and enrollment date).
- REQ-LC-03: A Creator shall be able to ban a specific learner from any of their own paths. The ban action shall immediately revoke the learner's access to the path's module content and progress tracking.
- REQ-LC-04: When a creator bans a learner, the ban is recorded in the system with the path, the banned user, the reason, and the date. The banned learner's enrollment status is updated to 'banned'.
- REQ-LC-05: A banned learner shall be able to view a notification that their access to the specific path has been revoked, but shall retain their account and access to all other enrolled paths.
- REQ-LC-06: A Creator shall be able to view their list of banned learners per path and optionally unban a learner, restoring their enrollment status to 'active'.
- REQ-LC-07: Administrators shall have read access to all ban records across all paths for audit purposes.

3.4 Course Enrollment & Progress Tracking (Learner)

3.4.1 Description and Priority

The Enrollment and Progress Tracking module enables authenticated users to enrol in any published career path and systematically track completion of each learning phase and module. Priority: High.

3.4.2 Functional Requirements

- REQ-EN-01: Authenticated users shall be able to enrol in any published career path via the enrollment interface (/user/enroll). Enrollment shall be blocked if the user has been banned from that path by its creator.
- REQ-EN-02: Upon enrollment, the system shall initialise a progress record for the learner tracking completion status for each module of the enrolled path.

- REQ-EN-03: The progress page (/user/progress) shall display the enrolled path, sequential phases and modules, and the completion status of each item with clear visual indicators.
- REQ-EN-04: Learners shall be able to mark individual modules as complete, with the system updating their progress record accordingly.
- REQ-EN-05: The POST /user/progress endpoint shall process progress updates submitted with a path title parameter and redirect to the progress GET view.
- REQ-EN-06: The User Home dashboard (/user/UserHome) shall display a summary of enrolled paths, progress metrics, and any new notifications from creators or admins.

3.5 Module Content Delivery

3.5.1 Description and Priority

This module delivers individual course content to learners through a dedicated module view, enabling structured consumption of learning materials phase by phase. Priority: High.

3.5.2 Functional Requirements

- REQ-MC-01: Authenticated learners shall be able to access individual module content via the /user/module endpoint, passing phase and title parameters to identify the specific module.
- REQ-MC-02: Before serving module content, the system shall verify that the requesting user is enrolled in the parent path and is not banned from it. Banned or non-enrolled users shall be redirected with an appropriate message.
- REQ-MC-03: The module content page shall present the module title, phase context, and associated learning content in a clear, readable layout with navigation to the next module and back to the progress overview.
- REQ-MC-04: The course detail view (/user/course_detail) shall present a full authenticated overview of a selected career path, including all phases, modules, and available resources, with an enrol button.
- REQ-MC-05: Guest users shall be able to access a preview course detail view (/guest/course_detail) without authentication, but shall be prompted to register to access full module content.

3.6 Certificate Generation

3.6.1 Description and Priority

The Certificate Generation module awards learners with verifiable completion certificates upon successfully finishing all modules and phases of an enrolled career path. Priority: Medium.

3.6.2 Functional Requirements

- REQ-CE-01: The system shall automatically evaluate whether a learner has completed all phases and modules of an enrolled career path and mark the path as completed.
- REQ-CE-02: Upon path completion, a certificate record shall be generated and made accessible from the Certificates page (/user/certificates).
- REQ-CE-03: The certificates page shall display all earned certificates, showing the path name, completion date, creator name, and a unique certificate identifier.

- REQ-CE-04: Certificates shall be available for download in PDF or printable HTML format.
- REQ-CE-05: Certificates shall not be accessible to learners who have not completed all required modules, or to learners who have been banned from the path before completing it.

3.7 Admin Dashboard & Platform Management

3.7.1 Description and Priority

The Admin Dashboard provides administrators with a centralised control panel for managing all aspects of the PathPilot system — users, all career paths regardless of creator, platform content, and notifications. Priority: High.

3.7.2 Functional Requirements

- REQ-AD-01: The dashboard home (/admin/dashboard) shall display platform-wide summary statistics: Total Users, Total Published Career Paths, Total Enrollments, and Recent Activity feed.
- REQ-AD-02: Administrators shall be able to view all registered users in a table with Name, Email, Role, Enrollment Count, Join Date, and Status columns; and edit roles, suspend, or delete any account.
- REQ-AD-03: Administrators shall be able to create new career paths (/admin/create-path), edit any existing path (/admin/edit-path), and delete any path, regardless of creator ownership.
- REQ-AD-04: Administrators shall be able to manage dynamic content for the platform's public-facing pages: Home (/admin/manage-home), Features (/admin/manage-features), About (/admin/manage-about), Contact (/admin/manage-contact), and Footer (/admin/manage-footer).
- REQ-AD-05: Administrators shall be able to broadcast notifications to all platform users via the Notify interface (/admin/notify).
- REQ-AD-06: Administrators shall have read-only access to all Creator-level ban records across all paths for audit and appeal purposes. Admins shall also be able to override bans.
- REQ-AD-07: All admin routes shall enforce server-side session validation (isNotAdmin check); unauthenticated or non-admin requests shall be redirected to /login.

4. External Interface Requirements

4.1 User Interfaces

The user interfaces of PathPilot are designed to deliver a seamless, modern, and visually consistent experience across all user roles and device types. All interfaces are implemented with Tailwind CSS to ensure full responsiveness. The colour palette is anchored by primary purple (#4913ec). The following screens constitute the complete UI specification:

- Index / Entry Point: index.jsp redirects all visitors to /guest/home.
- Guest Home: Public landing page with hero section, career path highlights, platform feature overview, and navigation to Home, Features, Resources, About, Career Path, and Contact.
- Guest Career Path Page: Browsable listing of all published career paths with phase count and description previews; prompts guests to register to enrol.
- Guest Course Detail Page: Preview of a selected path — phases, modules, and resources — with a registration prompt for full access.
- Login Page: Centred card with Email, Password, submit button, and register navigation link. Displays error messages for unverified accounts or invalid credentials.
- Registration Page: Centred card with Full Name, Email, and Password fields. On success, verification email is dispatched.
- User Home (Learner Dashboard): Personalised dashboard showing enrolled paths, progress summaries, creator analytics summary (paths authored, total learners), and recent notifications.
- User Career Path Page: Authenticated browsable listing with enrol button. Paths from which the user is banned display an access-revoked indicator.
- User Progress Page: Sequential phase and module checklist with completion indicators and View/Continue controls for each module.
- User Module Page: Module content delivery page parameterised by phase and title.
- Creator Path Management Page: Lists all paths authored by the logged-in user with learner count badge, Edit, Delete, and Manage Learners action buttons.
- Creator Learner Analytics Page: Per-path enrolled learner list with ban action button and unban control.
- User Certificates Page: Lists all earned certificates with path name, creator, date, and download option.
- User Profile & Settings Pages: Personal account management for name, bio, avatar, and account preferences.
- Admin Dashboard: Sidebar (Dashboard, Users, Career Paths, Platform Content, Notify, Settings) with statistics cards, ban audit log, and activity feed.
- Admin User Management: Add user (/admin/add-user) and Edit user (/admin/edit-users) forms for full account lifecycle management.
- Admin Career Path Management: Path listing, Create path, and Edit path interfaces with full phase/module editing capability.
- Admin Platform Content Pages: Separate CMS-style management pages for Home, Features, About, Contact, and Footer sections.
- Admin Notify Page: Compose and broadcast notification messages to all platform users.

4.2 Hardware Interfaces

- Client Devices: Accessible from any device with a modern web browser. Minimum recommended specification: 1 GHz processor, 512 MB RAM, screen resolution 360 px width minimum.
- Production Server: Minimum 2 GB RAM, 20 GB SSD storage, stable internet connection with sufficient bandwidth for concurrent users.
- Network: HTTPS for all client-server communication; outbound port 587 (TLS) for SMTP email dispatch.

4.3 Software Interfaces

- MySQL 8.0: Accessed via Spring JdbcTemplate for all CRUD operations on Users, CareerPaths, Phases, Modules, Enrollments, Progress, PathBans, and Certificates tables.
- Jakarta Mail 2.0.2 (SMTP): Used by EmailService to dispatch UUID token-based verification links. Credentials stored in server-side configuration only.
- Apache Tomcat 10.x: Spring MVC WAR deployed via AppInitializer (AbstractAnnotationConfigDispatcherServletInitializer) — no web.xml required.
- MySQL Connector/J 8.3.0: JDBC driver used by Spring JdbcTemplate.
- Tailwind CSS CDN: Responsive utility-based styling loaded via CDN with forms and container-queries plugins.
- Google Fonts CDN: Inter, Plus Jakarta Sans, Poppins typefaces and Material Icons Round loaded for all interface pages.
- Font Awesome CDN: Additional icon support on select admin and creator management pages.

4.4 Communications Interfaces

- HTTP/HTTPS Protocol: All client-server communication shall use HTTPS exclusively in production (TLS 1.2+). HTTP redirected to HTTPS.
- Spring MVC Request Handling: @GetMapping and @PostMapping annotated controller methods handle GET and POST HTTP requests. Form submissions use application/x-www-form-urlencoded encoding.
- Session Management: User state maintained via server-side Spring HttpSession (userEmail, role attributes). Validated per request via isNotUser / isNotAdmin helper methods.
- Email Communication: Transactional verification emails dispatched via SMTP over TLS on port 587 using Jakarta Mail through the EmailService Spring component.
- Ban Check: Before allowing a learner to access module content or re-enrol, the system checks whether that learner has been banned from the path by its creator.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- REQ-PF-01: All public-facing pages shall load within 3 seconds under normal network conditions with fewer than 500 concurrent users.
- REQ-PF-02: Career path listing, learner count queries, and progress queries shall return results within 2 seconds for a database containing up to 10,000 enrollment records.
- REQ-PF-03: Database queries shall be optimised to avoid slow full-table scans. Frequently searched fields shall be indexed to ensure fast response times.
- REQ-PF-04: Email verification link dispatch shall complete within 5 seconds of registration submission.
- REQ-PF-05: The system shall support a minimum of 200 concurrent active user sessions without measurable degradation in response time.
- REQ-PF-06: Learner count aggregation queries on the Creator Analytics view shall complete within 3 seconds even for paths with more than 1,000 enrolled learners.

5.2 Safety Requirements

- REQ-SA-01: All user passwords shall be stored using a secure one-way hashing algorithm. Plain-text password storage is strictly prohibited.
- REQ-SA-02: All data transmitted between client and server shall be encrypted using TLS 1.2 or higher in the production environment.
- REQ-SA-03: All user input shall be validated on both client side (HTML5 + JavaScript) and server side (Spring controller validation) before processing.
- REQ-SA-04: Email verification tokens shall be single-use UUID values. Once consumed, the token shall be set to NULL in the database.
- REQ-SA-05: Every ban action by a creator shall be recorded with the creator's identity, the banned learner's identity, the path, and the reason given. This log is available to administrators for review.
- REQ-SA-06: Role-based and path-level access controls shall be enforced exclusively server-side. Client-side restrictions alone are not relied upon.

5.3 Security Requirements

- REQ-SE-01: The system shall protect against unauthorised access by validating user sessions on every protected page. Unauthenticated requests shall be redirected to the login page.
- REQ-SE-02: All user passwords shall be stored securely and shall not be stored or transmitted in plain text.
- REQ-SE-03: The system shall ensure that a creator can only manage and moderate learners on paths they personally own. Attempting to modify another creator's path shall be blocked by the system.

5.4 Software Quality Attributes

- Usability: Navigation to any primary feature shall require no more than three clicks from any page. The interface shall conform to WCAG 2.1 Level AA accessibility guidelines.
- Reliability: Minimum uptime of 99.5% per calendar month. Core features (login, career path browsing, progress tracking, creator analytics) shall be continuously available during scheduled uptime windows.
- Scalability: The database design shall support growth to 50,000 registered users, 500 career paths, and 100,000 enrollment records without requiring a full rebuild of the system.
- Maintainability: Java backend organised into logical Spring MVC layers (controllers, services, config). All non-trivial methods shall include Javadoc comments. Complete ER diagram maintained as technical documentation.
- Portability: Deployable on any server running Java 21+, Tomcat 10.x, and MySQL 8.0 without code modification.
- Testability: Critical controller and service methods accompanied by JUnit tests. Database interaction modules support integration testing. UAT plan executed with representative learner and creator users before launch.

6. Other Requirements

- Legal and Compliance: Terms of Service and Privacy Policy must be explicitly accepted at registration. The platform shall comply with applicable data protection regulations.
- Content Policy Enforcement: Creator-published career paths and resource materials must comply with the platform's published content policy. Administrators have authority to remove non-compliant content regardless of creator.
- Creator Community Standards: The Creator ban feature shall be governed by published community standards. Creators may not abuse the ban feature to discriminate against learners. Admins can review and override any ban.
- Localisation: Initial release in English. Architecture shall support future internationalisation without significant code refactoring.
- Data Backup and Recovery: Automated daily MySQL database backups in production. Recovery procedures documented and tested quarterly.
- Monitoring and Alerting: Administrators shall have access to real-time server health metrics. Automated alerts configured for threshold breaches.
- Notification System: Admin broadcast notifications to all users. Creator-to-learner notifications (e.g., path update announcements) shall be supported in a future version.
- SEO Optimisation: All publicly accessible pages shall include appropriate HTML meta tags (title, description, Open Graph) for search engine indexing.
- Testing Requirements: Basic testing of all key features (login, enrollment, path creation, banning) and testing with real student users shall be completed before the project is submitted for evaluation.

Appendix A: Glossary

- PathPilot: A web-based career guidance and learning platform where any registered user can simultaneously learn from published career paths and create their own paths for others to enrol in.
- SRS (Software Requirements Specification): A formal document describing the intended purpose, environment, features, and constraints of a software system.
- RBAC (Role-Based Access Control): Security model restricting system access based on assigned roles (Guest/unauthenticated, USER, ADMIN).
- Guest: An unauthenticated visitor who can browse public content but cannot enrol, create paths, or access personalised features.
- Learner: A logged-in user in their capacity as a student — enrolling in career paths created by others and tracking their progress.
- Creator: A logged-in user in their capacity as a content publisher — building career paths, monitoring learner counts, and moderating their learner community.
- Learner/Creator: PathPilot's unified logged-in user role; every registered user is both a Learner and a Creator simultaneously.
- Career Path: A structured, multi-phase learning journey built by a Creator (or Admin), containing sequential phases and modules enrollable by any user.
- PathBan: A record in the PathBans table representing a Creator's decision to revoke a specific learner's access to one of their owned paths.
- Enrollment: A persistent record linking a Learner to a Career Path, tracking the learner's engagement and progress status (active, completed, banned).
- Spring MVC: Java web framework following the Model-View-Controller pattern used as the PathPilot backend.
- JdbcTemplate: A Spring helper class that simplifies communication between the Java application and the MySQL database.
- WAR (Web Application Archive): Packaged Java web application deployed to Apache Tomcat.
- JSP (Jakarta Server Pages): Server-side templating technology generating dynamic HTML views.
- UUID (Universally Unique Identifier): 128-bit identifier used as email verification token.
- SMTP (Simple Mail Transfer Protocol): Protocol used by EmailService to dispatch verification emails.
- TLS (Transport Layer Security): Cryptographic protocol for secure data transmission over HTTPS and SMTP.

Appendix B: Analysis Models

This appendix contains the analysis and design models developed for the PathPilot system. These models document the system architecture, use case relationships, class structure, database entities, and inter-component relationships, providing a comprehensive reference for design and implementation activities.

Use Case Diagram

The Use Case Diagram below illustrates the three user roles — Guest, Learner/Creator, and Admin — and the complete set of system interactions available to each role. The Learner/Creator role inherits all Guest capabilities and additionally encompasses both learning (enrolment, progress, certificates) and content creation (path creation, learner analytics, learner banning).

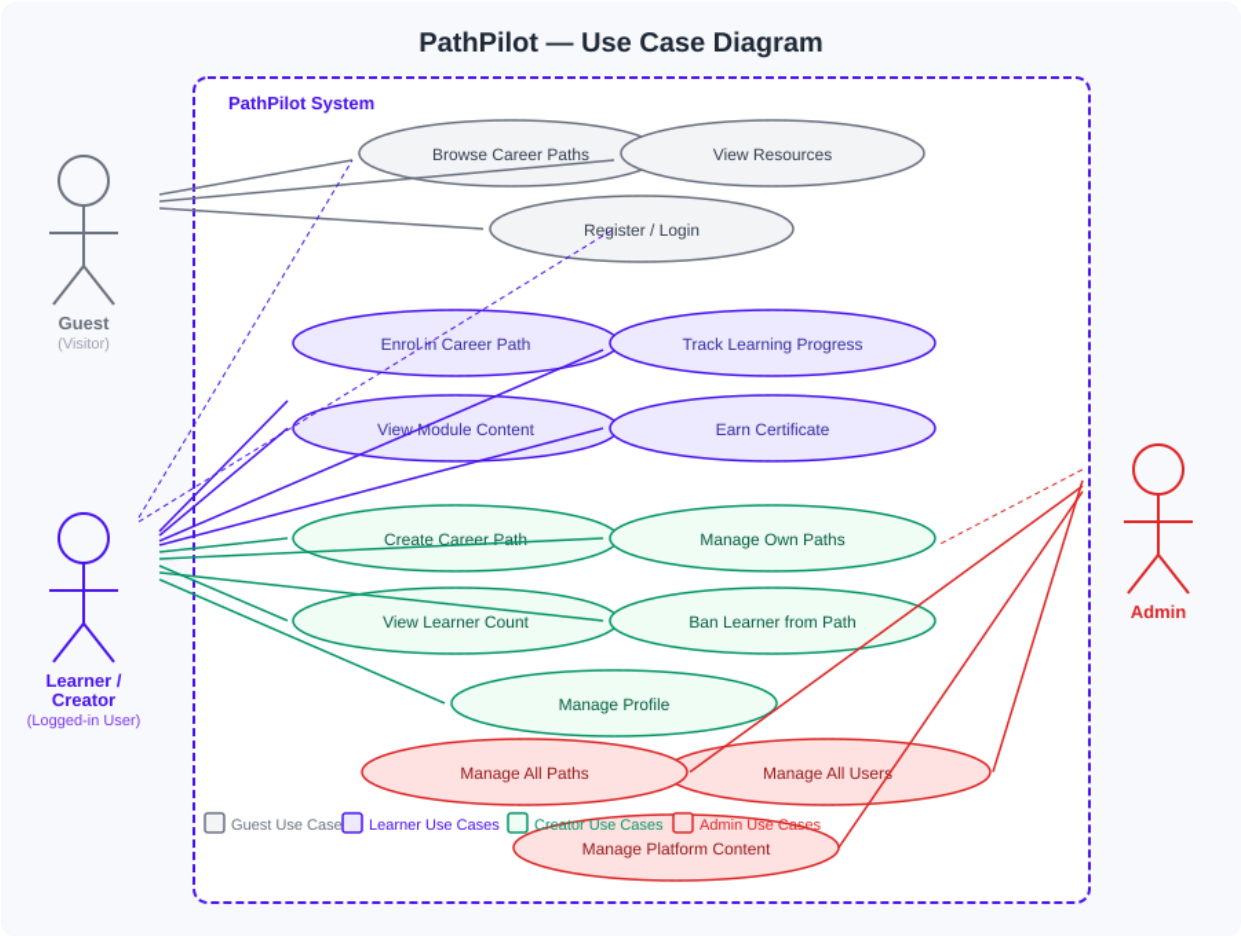

Figure 1: PathPilot — Use Case Diagram showing Guest, Learner/Creator, and Admin roles

Class Diagram

The Class Diagram below presents the core classes of the PathPilot system, their attributes, methods, and relationships. The User base class is extended by Guest, Learner/Creator, and Admin subclasses. The Learner/Creator class holds all creator-specific methods (createCareerPath, viewLearnerCount, banLearner) in addition to all learner-specific methods. The CareerPath, Enrollment, and Certificate classes capture the key data entities and their associations.

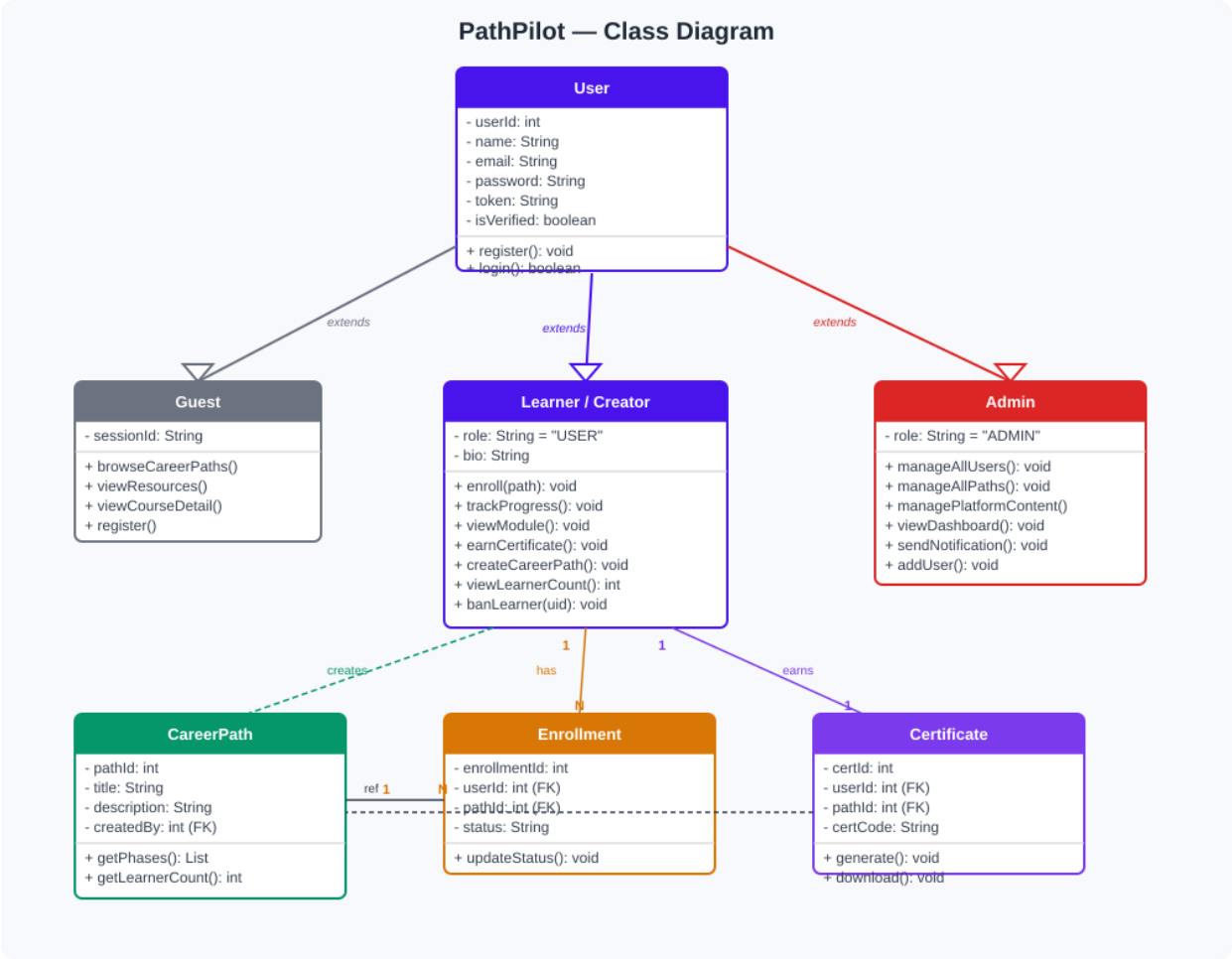

Figure 2: PathPilot — Class Diagram showing inheritance hierarchy and entity associations

Database Overview

The PathPilot database stores all platform data across the following main tables: Users, CareerPaths, Phases, Modules, Enrollments, Progress, PathBans, and Certificates. Each user can create many career paths and enrol in many paths. Each career path contains multiple phases, and each phase contains multiple modules. When a learner completes all modules in a path, a certificate is generated. The PathBans table records any learner bans applied by creators on their own paths.

Appendix C: Issues List

The following table identifies known challenges, risks, and open issues associated with the PathPilot development project. Each item is assigned a priority level to guide mitigation planning.

#	Issue / Risk	Mitigation Strategy	Priority
1	Passwords stored in plain text in prototype	Store all passwords using a secure hashing method before going live.	Critical
2	Unauthorised access to other creator's paths	The system must check that the logged-in user owns the path before allowing any edit, delete, or ban action.	Critical
3	Email verification email not delivered	Show a 'Resend Verification Email' option on the login page if the user's account is not yet verified.	High
4	Ban feature misused by creators	Log every ban with a reason and date. Admins can review and reverse any ban.	High
5	Slow page loading with many enrolled users	Optimise database queries and add pagination to listing pages.	Medium
6	No automated database backup	Set up automatic daily backups on the production server.	High
7	Tailwind CSS loaded from internet (CDN)	Host the CSS file locally on the server so the site works without internet dependency.	Medium
8	Site may slow down as more users join	Plan for server upgrade and use pagination on all list views from the start.	Low

Regular testing, user feedback collection, and continuous improvements are essential to managing the issues identified above. The development team commits to resolving all Critical and High priority items prior to the production launch of PathPilot.